(assistance under GPT5.6)

3+3+(2) scaling laws for LLM Pre-training (via random matrix theory)

Fanghui Liu

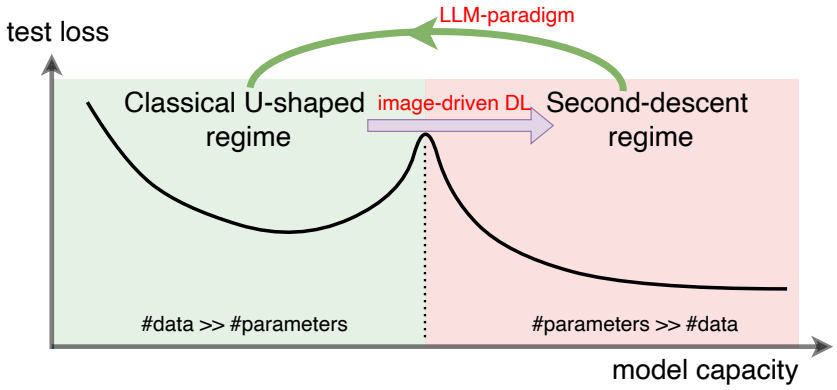
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Generalization under scaling



- U-shaped curve (Vapnik, 1995; Hastie et al., 2009)
- Double descent (Belkin et al., 2019, Nakkiran et al., 2019)
- Scaling law (Kaplan et al., 2020; Hoffmann et al., 2022)

Compute-optimal scaling laws

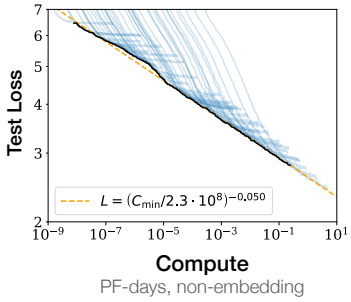


Figure 1: OpenAI / Kaplan et al. (2020)

Scaling laws (e.g., Kaplan, Chinchilla)

$$\mathcal{L}(M, N) \simeq \mathcal{L}_\infty + aM^{-\alpha} + bN^{-\beta}$$

M : model size; N : training token

Method	OpenAI	DeepMind
$M_\star$	$C^{0.73}$	$C^{0.49}$
$N_\star$	$C^{0.27}$	$C^{0.51}$
TPP $N_\star/M_\star$	$C^{-0.46}$	constant
Policy	$\nearrow M \gg \nearrow N$	$\nearrow M \approx \nearrow N$

Is a scaling law universal?

Origin

When does a learning curve follow a power law?

Transfer

If a power law appears, does its exponent survive a change of schedule?

Kaplan et al. (2020); Hoffmann et al. (2022)

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Origin

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If a power law appears, does its exponent survive a change of schedule?

1

Existence

What are the necessary and sufficient conditions for a power-law response?

2

Modeling

Which macroscopic quantities turn spectrum, target, and schedule into loss?

3

Prediction

Can a response learned on one schedule predict a held-out LLM schedule?

Kaplan et al. (2020); Hoffmann et al. (2022)

A technical science view

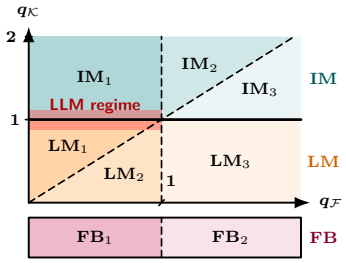
Question	Concept
Can a proxy model capture the mechanisms of LLMs? Can macroscopic quantities capture loss?	**Modeling (Analysis)**
Is the loss predictable?	**Input-output prediction**
Can loss or gradient histories distinguish states with different responses to a future schedule change?	**Observability**
Does a schedule achieve the target loss $R_N = R_{\mathrm{tar}}$?	**Output reachability (Design)**
Does a schedule reach the target state $s_N = s_{\mathrm{tar}}$?	**State reachability**

A technical science view

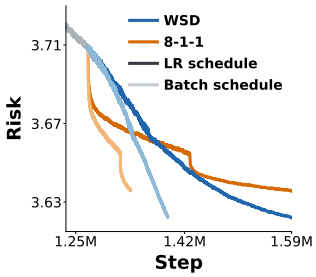
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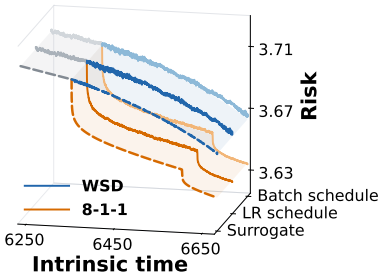
Empirical validation



(a) $3 + 3(+2)$ map



(b) Loss vs. step

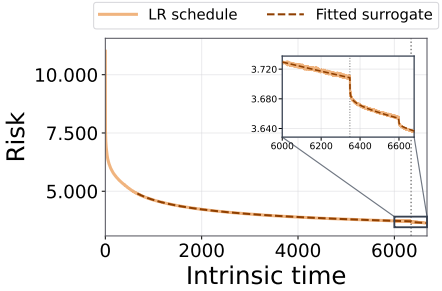


(c) Loss vs. intrinsic time

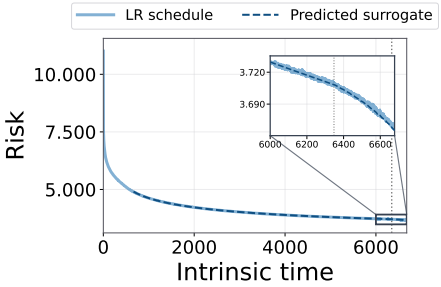
Empirical prediction (a re-parameterized functional scaling laws)

(Inspired by Li et al. Functional scaling laws. NeurIPS'25.)

- Consider a 7-parameter model (derived from our theory)
- curve fitting on 8-1-1 schedules
- predict the loss curve under WSD



(a) 8-1-1 fit



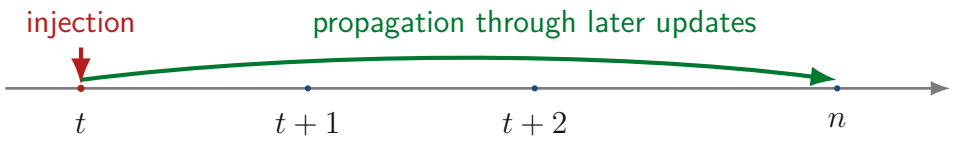
(b) WSD prediction

SGD: one noise injection can affect future loss

One stochastic update

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta_t \nabla \mathcal{L}(\boldsymbol{\theta}_t) - \eta_t \boldsymbol{\xi}_t, \qquad \mathrm{Cov}_t(\boldsymbol{\xi}_t) = \frac{\boldsymbol{\Sigma}(\boldsymbol{\theta}_t)}{B_t}.$$

For nonlinear models, this response is a small-noise approximation.



What one injection contributes

- **Variance of injection:** $\mathrm{Cov}_t(-\eta_t \boldsymbol{\xi}_t) = (\eta_t^2 / B_t)\boldsymbol{\Sigma}(\boldsymbol{\theta}_t)$.
- **Accumulated history:** Present loss depends on the full history of injections.

Measure progress, then noise per unit progress

1. Why intrinsic time?

(Li et al. Functional scaling laws. NeurIPS'25.)

Step count ignores differences in the learning rate.

$$\Delta T_s = \eta_s, \qquad T_t = \sum_{s<t} \eta_s.$$

Equal values of T represent equal first-order optimization progress.

2. Why the ratio?

Per-step noise variance scales as η_s^2/B_s. Per unit intrinsic time,

$$\frac{\eta_s^2/B_s}{\Delta T_s} = \frac{\eta_s}{B_s} = \frac{1}{r_s}, \qquad r_s := \frac{B_s}{\eta_s}.$$

Hence r_s is the inverse noise intensity per unit progress.

different update counts; same total intrinsic time



Two schedule coordinates: T measures optimization progress, while $r(T)$ controls the accumulation of stochastic error.

Theory of proxy model (iff criterion)

A proxy that isolates the mechanism

Frozen random features

$$f_{\boldsymbol{a}}(\boldsymbol{x}) = \langle \boldsymbol{W}^\top \boldsymbol{x}, \boldsymbol{a} \rangle, \qquad y = \langle \boldsymbol{x}, \boldsymbol{\theta}^\star \rangle + \varepsilon.$$

- $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}_d)$ and $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_d) \succ \boldsymbol{0}$
- $\boldsymbol{W} \in \mathbb{R}^{d \times m}$ is fixed with $\mathcal{N}(0, \frac{1}{m})$.
- Only $\boldsymbol{a}$ is trained.

Noisy online SGD

$$\boldsymbol{a}_{t+1} = \boldsymbol{a}_t - \frac{\eta_t}{B_t} \sum_{i=1}^{B_t} \boldsymbol{W}^\top \boldsymbol{x}_t^i \big(f_{\boldsymbol{a}_t}(\boldsymbol{x}_t^i) - y_t^i \big).$$

Rahimi and Recht (2007); Liu et al. (2021); Mei et al. (2022); Paquette et al. (2024)

response mechanism

vs.

literal equivalence

Why this proxy?

It separates four effects that evolve together in an LLM:

- spectrum and representation;
- target alignment;
- stochastic injection;
- joint scheduling.

From prediction risk to spectral filtering (extension of Paquette et al. (2024))

1. **Decompose the risk into spectral modes.**

$$\widehat{\boldsymbol{H}} := \boldsymbol{\Lambda}^{1/2}\boldsymbol{W}\boldsymbol{W}^{\top}\boldsymbol{\Lambda}^{1/2}, \qquad \widehat{\boldsymbol{H}}\widehat{\boldsymbol{u}}_j = \widehat{\lambda}_j\widehat{\boldsymbol{u}}_j,$$

$$\boldsymbol{e}_t := \boldsymbol{\Lambda}^{1/2}(\boldsymbol{W}\boldsymbol{a}_t - \boldsymbol{\theta}^{\star}), \qquad R_{\sigma,t} := \mathbb{E}[\|\boldsymbol{e}_t\|_2^2|\boldsymbol{W}] = \sum_j \underbrace{\mathbb{E}\big|\langle\widehat{\boldsymbol{u}}_j, \boldsymbol{e}_t\rangle\big|^2}_{\mathcal{E}_{j,t}},$$

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2. Propagate each modal energy by one SGD step. For constant (η, B),

$$\mathcal{E}_{j,t+1} = \underbrace{q_\eta(\widehat{\lambda}_j)\mathcal{E}_{j,t}}_{\text{free propagation of old error}} + \underbrace{\frac{\eta^2}{B}\widehat{\lambda}_j^2\big(R_{\sigma,t} + \sigma^2\big)}_{\text{feedback and noise injection}}$$

$$q_\eta(\lambda) = \underbrace{(1-\eta\lambda)^2}_{\text{full-batch contraction}} + \underbrace{\frac{\eta^2\lambda^2}{B}}_{\text{mini-batch fluctuation}}$$

Volterra equation

$$R_{\sigma,t} = \underbrace{F_{\boldsymbol{W}}(t)}_{\text{unresolved target error}} + \sum_{s=0}^{t-1} \underbrace{K_{\boldsymbol{W}}(t-1-s)}_{\text{survival of one injection}} \Big(\underbrace{R_{\sigma,s}}_{\text{feedback}} + \underbrace{\sigma^2}_{\text{label noise}} \Big).$$

Forcing F
Propagates the initial target error without future stochastic feedback.

Memory K
The impulse response of a single SGD-variance injection.

Observed loss R_σ
Accumulates all past injections and feeds earlier risk back into the dynamics.

For eigenpairs $(\widehat{\lambda}_j, \widehat{\boldsymbol{u}}_j)$,

$$F_{\boldsymbol{W}}(t) = \sum_j \left| \left\langle \widehat{\boldsymbol{u}}_j, \boldsymbol{\Lambda}^{1/2} \boldsymbol{\theta}^\star \right\rangle \right|^2 q_\eta(\widehat{\lambda}_j)^t, \qquad K_{\boldsymbol{W}}(t) = \frac{\eta^2}{B} \sum_j \widehat{\lambda}_j^2 q_\eta(\widehat{\lambda}_j)^t.$$

The power law of F or K alone is not yet a power law for the total loss.

SGD: bias-variance decomposition (Wu et al. 2022, Liu et al. 2022)

Online linear regression: fresh i.i.d. samples and a fixed initialization $\boldsymbol{w}_0$.

$$y_t = \boldsymbol{x}_t^\top \boldsymbol{w}^\star + \varepsilon_t, \quad \boldsymbol{H} = \mathbb{E}[\boldsymbol{x}\boldsymbol{x}^\top], \quad \varepsilon_t \perp \boldsymbol{x}_t, \quad \mathbb{E}[\varepsilon_t] = 0, \quad \mathbb{E}[\varepsilon_t^2] = \sigma^2.$$

Let $\boldsymbol{e}_t = \boldsymbol{w}_t - \boldsymbol{w}^\star$. Split the same SGD trajectory by error source:

$$\boldsymbol{e}_t = \boldsymbol{u}_t + \boldsymbol{z}_t, \qquad \boldsymbol{A}_t = \boldsymbol{I} - \gamma_t \boldsymbol{x}_t \boldsymbol{x}_t^\top.$$

Bias process: initial error

$$\boldsymbol{u}_t = \boldsymbol{A}_t \boldsymbol{u}_{t-1}, \quad \boldsymbol{u}_0 = \boldsymbol{e}_0,$$
$$b_t = \mathbb{E}\|\boldsymbol{u}_t\|_{\boldsymbol{H}}^2.$$

No label noise; random sample updates.

Variance process: label noise

$$\boldsymbol{z}_t = \boldsymbol{A}_t \boldsymbol{z}_{t-1} + \gamma_t \varepsilon_t \boldsymbol{x}_t,$$
$$\boldsymbol{z}_0 = \boldsymbol{0}, \qquad v_t = \mathbb{E}\|\boldsymbol{z}_t\|_{\boldsymbol{H}}^2.$$

Start at $\boldsymbol{w}^\star$; retain label noise.

The cross term $\mathbb{E}[\boldsymbol{u}_t^\top \boldsymbol{H} \boldsymbol{z}_t] = 0$, hence

$$\boxed{R_t := \mathbb{E}\|\boldsymbol{e}_t\|_{\boldsymbol{H}}^2 = b_t + v_t.}$$

Here, bias includes sampling fluctuations during signal learning.

$$b_t = \|\mathbb{E}\boldsymbol{e}_t\|_{\boldsymbol{H}}^2 + \mathrm{Tr}\big(\boldsymbol{H}\,\mathrm{Cov}(\boldsymbol{u}_t)\big).$$

Relations between forcing-memory and bias-variance

For $\boldsymbol{x} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{H})$ and varying stepsizes, the exact discrete recursion is

$$R_t = F_t + M_t, \qquad M_t := \sum_{s=1}^{t} \mathcal{K}_{t,s}(R_{s-1} + \sigma^2), \qquad R_t = b_t + v_t.$$

Volterra component	Bias–variance interpretation
F_t	Initial-error propagation within bias
$\sum_{s=1}^{t} \mathcal{K}_{t,s} b_{s-1}$	**Sampling feedback within bias**
$\sigma^2 \sum_{s=1}^{t} \mathcal{K}_{t,s}$	Direct injection and propagation of label noise
$\sum_{s=1}^{t} \mathcal{K}_{t,s} v_{s-1}$	Sampling feedback from existing noise error

$$b_t = F_t + \sum_{s} \mathcal{K}_{t,s} b_{s-1}, \qquad \boxed{M_t = (b_t - F_t) + v_t.}$$

Memory includes bias feedback and the full variance. It can be nonzero even if $\sigma^2 = 0$.

F_t includes a discrete sampling correction; generally $F_t \neq \|\mathbb{E}\boldsymbol{e}_t\|_{\boldsymbol{H}}^2$.

Two responses, two spectral weights

Forcing: target-weighted spectrum

$$\nu^{\mathcal{F}}_{\boldsymbol{W}}((0,x]) = \sum_{0<\widehat{\lambda}_j \le x} \left|\left\langle \widehat{\boldsymbol{u}}_j, \boldsymbol{\Lambda}^{1/2}\boldsymbol{\theta}^\star \right\rangle\right|^2.$$

It measures the target error placed in slow directions $0 < \widehat{\lambda}_j \le x$.

$$F_{\boldsymbol{W},>0}(t) = \int_0^\infty q_\eta(\lambda)^t \, \nu^{\mathcal{F}}_{\boldsymbol{W}}(\mathrm{d}\lambda).$$

Memory: squared-spectrum weight

$$\nu^{\mathcal{K}}_{\boldsymbol{W}}((0,x]) = \sum_{0<\widehat{\lambda}_j \le x} \widehat{\lambda}_j^2.$$

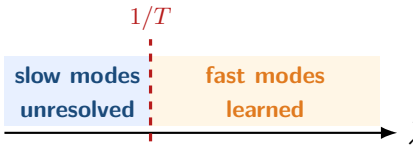
It measures how strongly one SGD-noise injection is carried by these slow directions.

$$\frac{B}{\eta^2} K_{\boldsymbol{W}}(t) = \int_0^\infty q_\eta(\lambda)^t \, \nu^{\mathcal{K}}_{\boldsymbol{W}}(\mathrm{d}\lambda).$$

The same dynamics filters two different spectral weights: target alignment for forcing, and squared eigenvalues for memory.

Weighted spectral tails characterize power laws (iff criterion)

$$q_\eta(\lambda)^t \approx e^{-2T\lambda}, \qquad T := \eta t, \qquad e^{-2T\lambda} \approx \mathbf{1}\{\lambda \lesssim T^{-1}\}\,.$$



The exponential filter probes cumulative spectral mass near the moving cutoff $x \asymp T^{-1}$:

$$\int_{(0,\infty)} e^{-2T\lambda}\nu(\mathrm{d}\lambda) = 2T\int_0^\infty e^{-2Tx}\nu((0,x])\,\mathrm{d}x.$$

Thus, if $\nu((0,x]) \propto x^q$ near zero, the response is of order T^{-q}.

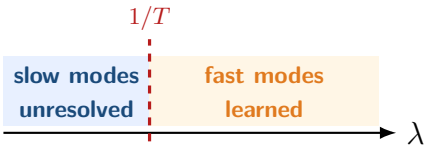
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Componentwise spectral criterion (informal)

Under the stable joint width-time limit,

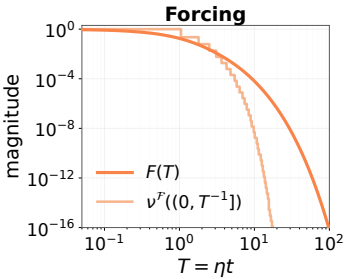
$$\nu^{\mathcal{F}}_{\boldsymbol{W}}((0,x]) \propto x^{q_{\mathcal{F}}} \iff F_{\boldsymbol{W},>0}(t) \propto T^{-q_{\mathcal{F}}},$$

$$\nu^{\mathcal{K}}_{\boldsymbol{W}}((0,x]) \propto x^{q_{\mathcal{K}}} \iff \frac{B}{\eta^2}K_{\boldsymbol{W}}(t) \propto T^{-q_{\mathcal{K}}}, \qquad x = T^{-1} \downarrow 0.$$

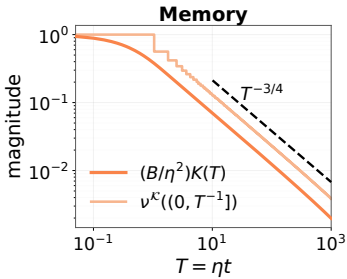
Power-law data (Zipf laws) are neither sufficient nor necessary

The spectrum has slow directions, but the target barely uses them.

$$\lambda_j = j^{-0.8}, \qquad |\theta_j^\star|^2 = e^{-j}.$$

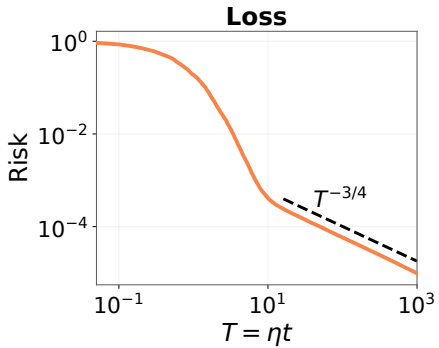


Forcing: faster than every inverse power.



Memory: $T^{-3/4}$.

A total-loss exponent does not identify its mechanism



What sets the tail?

- **Forcing.** Faster than every power law.
- **Memory.** Decays as $T^{-3/4}$.
- **Late SGD loss.** Follows the memory tail.

A fitted loss exponent does not identify what is still being learned.

The loss curve is many-to-one: it cannot reveal the underlying mechanism by itself.

Power-law random features model (regimes)

The canonical PLRF model makes the cutoff calculation explicit

Assume the microscopic power laws

$$\lambda_j = j^{-2\alpha}, \qquad \theta_j^\star = j^{-\beta}, \qquad 2\alpha + 2\beta > 1.$$

The moving cutoff $\lambda_{j_T} \asymp T^{-1}$ gives $j_T \asymp T^{1/(2\alpha)}$. Summing the unresolved tail yields

$$F(T) \approx \sum_{j \gtrsim j_T} j^{-2(\alpha+\beta)} \propto T^{-q_{\mathcal{F}}}$$

$$\boxed{q_{\mathcal{F}} = \frac{2\alpha + 2\beta - 1}{2\alpha}}$$

Spectrum *and* target alignment

$$K(T) \approx \sum_{j \gtrsim j_T} j^{-4\alpha} \propto T^{-q_{\mathcal{K}}}$$

$$\boxed{q_{\mathcal{K}} = 2 - \frac{1}{2\alpha}}$$

Spectrum alone

The condition $2\alpha + 2\beta > 1$ ensures finite target energy: $\sum_j \lambda_j |\theta_j^\star|^2 < \infty$.

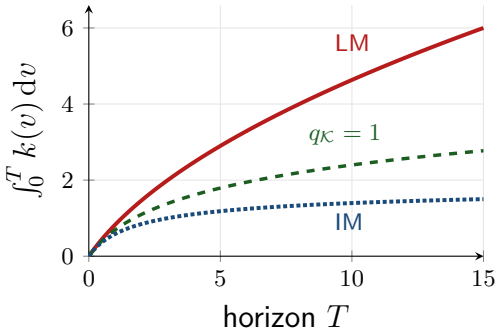
Why use $(q_{\mathcal{F}}, q_{\mathcal{K}})$, rather than only (α, β)?

	(α, β)	$(q_{\mathcal{F}}, q_{\mathcal{K}})$
Meaning	microscopic spectrum and source parameters	decay of target forcing and one-injection memory
Where defined	canonical coordinatewise PLRF assumptions	any dynamics with the corresponding response laws

The honest interpretation

Inside PLRF, this is an algebraic change of coordinates. Outside PLRF, irregular spectra and targets may have no meaningful (α, β), while $(q_{\mathcal{F}}, q_{\mathcal{K}})$ can still describe their macroscopic response.

Three memory regimes

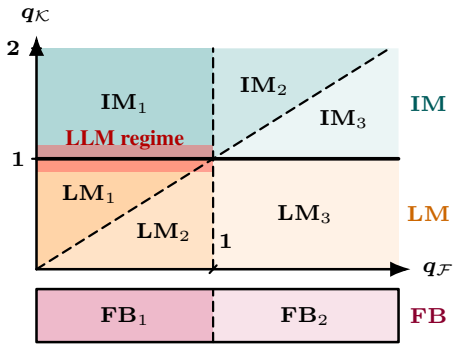


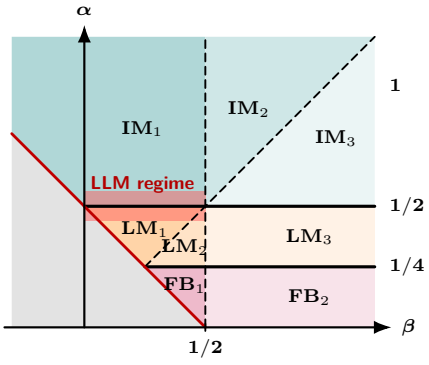
Three memory regimes

- **Long memory (LM):** $\mathbf{0 < q_{\mathcal{K}} < 1}$. Accumulated effect grows.
- **Marginal:** $\mathbf{q_{\mathcal{K}} = 1}$. $k(v) \sim c/v$: logarithmic growth.
- **Integrable memory (IM):** $\mathbf{q_{\mathcal{K}} > 1}$. Total memory is finite.

$\int_0^T k(v)\,\mathrm{d}v$ sums surviving injections: divergence gives LM; convergence gives IM.

The $3 + 3(+2)$ scaling law regimes





- **Long memory (LM):** $0 < q_{\mathcal{K}} < 1$ $(\frac{1}{4} < \alpha < \frac{1}{2})$
  Memory is nonintegrable; the distant past remains important.
- **Integrable memory (IM):** $q_{\mathcal{K}} > 1$ $(\alpha > \frac{1}{4})$ Total memory is finite; recent injections dominate.
- **Finite bulk (FB)** $0 < \alpha < \frac{1}{4}$ Memory stays explicitly coupled to width. $(\sum_{j \le m} \lambda_j^2 \asymp m^{1-4\alpha})$

What does random matrix theory do?

1. Encode the spectral dynamics by the resolvent.

Both forcing and memory are spectral functions of $\widehat{\boldsymbol{H}}$. For an analytic g,

$$\boldsymbol{G}_m(z) := (\widehat{\boldsymbol{H}} - z\boldsymbol{I})^{-1}, \qquad g(\widehat{\boldsymbol{H}}) = -\frac{1}{2\pi i} \oint_{\mathcal{C}} g(z)\boldsymbol{G}_m(z)\,\mathrm{d}z\,.$$

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2. Replace the random resolvent by a deterministic equivalent.

For the required bilinear and trace observables, RMT gives

$$\boldsymbol{G}_m(z) \;\rightsquigarrow\; \mathscr{R}_m\big(z; \mathfrak{m}(z)\big),$$

where one scalar fixed point $\mathfrak{m}(z)$ captures the effect of the random projection.

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3. Read the learning laws from low frequency.

Expanding the deterministic equivalent near $z = 0$ gives

$$z \longrightarrow 0 \quad \Longrightarrow \quad \big(R_{\mathrm{app}}, q_{\mathcal{F}}, q_{\mathcal{K}}\big) \quad \Longrightarrow \quad 3 + 3(+2) \text{ regimes.}$$

The low spectrum generates the power law. RMT computes how random features modify it.

Application: Scaling laws via joint schedules

Joint schedules reduce to two response coordinates

Optimization clock

$$T_t = \sum_{s<t} \eta_s$$

Each step advances the slow-mode filter by η_s.

Noise-control ratio

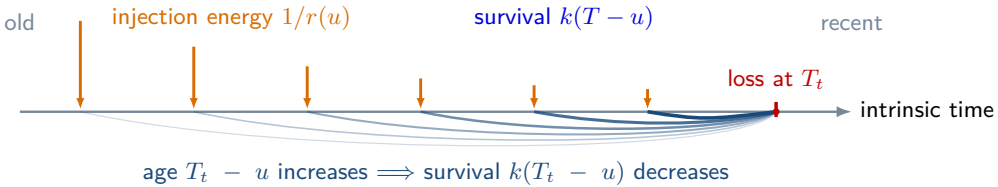
$$r_t = \frac{B_t}{\eta_t}$$

The variance injected per unit intrinsic time is proportional to $1/r_t$.

different (η_t, B_t) factorizations → same T and same path $r(T)$

The leading plain-SGD response predicts approximate factorization invariance in these two coordinates.

A schedule is an input; memory is the system response



Subtracting the clean recursion from the noisy one and passing to intrinsic time gives

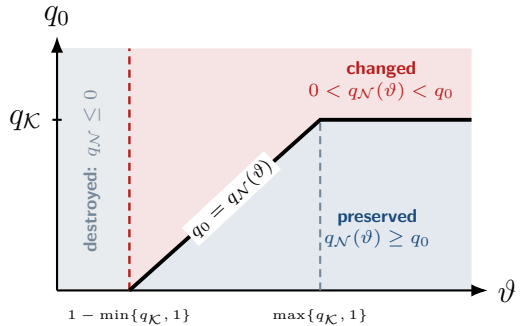
$$R_\sigma(T) \asymp F(T) + \int_0^T \underbrace{\frac{F(u)+\sigma^2}{r(u)}}_{\text{injected at } u} \underbrace{k(T-u)}_{\text{survives to } T} \, \mathrm{d}u.$$

The total noise response depends on the **entire history** of the ratio path, not only its terminal value.

A schedule can preserve, change, or destroy the clean scaling law

$$k(v) \sim v^{-q_{\mathcal{K}}}, \qquad r(T) \sim T^{\vartheta} \quad \Longrightarrow \quad \Delta(T) := R_\sigma(T) - R_0(T) \asymp T^{-q_{\mathcal{N}}(\vartheta)}.$$

$$q_{\mathcal{N}}(\vartheta) = \begin{cases} \min\{\vartheta + q_{\mathcal{K}} - 1, q_{\mathcal{K}}\}, & \mathrm{LM}(0 < q_{\mathcal{K}} < 1), \\ \min\{\vartheta, q_{\mathcal{K}}\}, & \mathrm{IM}(q_{\mathcal{K}} > 1), \end{cases}$$



Noise response and observed exponent

$$R_0(T) - R_{\mathrm{app}} \sim c_0 T^{-q_0},$$
$$R_\sigma(T) - R_{\mathrm{app}} \asymp T^{-q_{\mathrm{obs}}(\vartheta)},$$
$$q_{\mathrm{obs}}(\vartheta) = \min\{q_0, q_{\mathcal{N}}(\vartheta)\}.$$

At the boundary $q_{\mathcal{K}} = 1$, logarithmic corrections may appear.

The schedule response determines the optimal data law

Since $B_s = r(T_s)\eta_s$, the data consumed by a ratio schedule satisfies

$$D = \sum_{s<t} B_s = \sum_{s<t} r(T_s)\eta_s \approx \int_0^T r(u)\,\mathrm{d}u.$$

Thus stronger noise control also uses more data.

Balance unresolved signal and budget-limited noise

PLRF specialization: $\lambda_j = j^{-2\alpha}$ and $p := 2\alpha + 2\beta - 1 > 0$.

$$\mathcal{R}_\sigma(T; D) \asymp \underbrace{T^{-p/(2\alpha)}}_{\text{Unresolved signal}} + \underbrace{D^{-1}T^{1/(2\alpha)}}_{\text{Budget-limited noise}}.$$

Balancing them gives

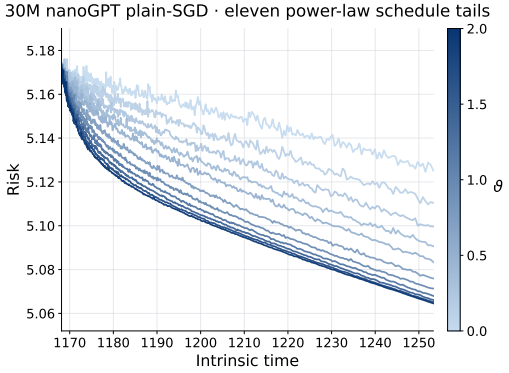
$$T^\star \asymp D^{2\alpha/(p+1)} \qquad \Longrightarrow \qquad \mathcal{R}_\sigma^\star(D) \asymp D^{-p/(p+1)}$$

Experiments

Intrinsic time is insufficient

30M / plain SGD / OpenWebText / fixed batch size $B = 8$; Risk = validation CE

- The same intrinsic time
- Different optimizer steps and tokens

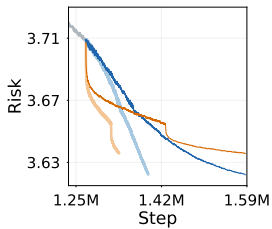


Matched paths decided by intrinsic time and ratio

300M / plain SGD / OWT / 6.5B tokens; Risk = validation CE

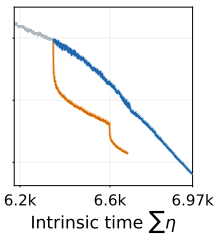
Optimizer step

Each step advances T by η.



Intrinsic time

Same $r(T)$; each pair overlaps.



WSD: LR (fixed B_t)

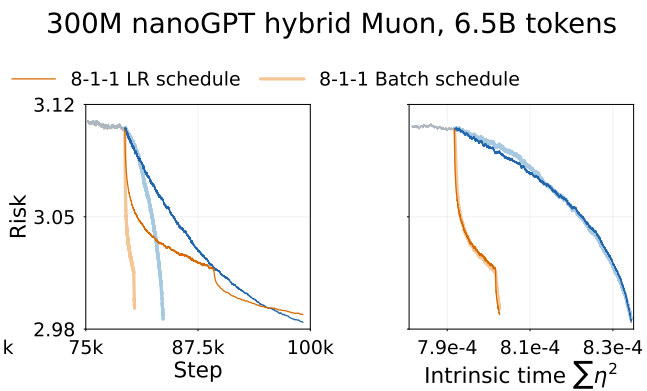
WSD: batch (fixed η_t)

8-1-1: LR (fixed B_t)

8-1-1: batch (fixed η_t)

Finding. Matched $r(T)$ makes LR/batch factorizations overlap in intrinsic time.

Go beyond SGD: Muon suggests optimizer-specific coordinates



A separate short calibration selects

$$\widetilde{T}_t = \sum_{s<t} \eta_s^2, \qquad \widetilde{r}_t = \frac{B_t}{\eta_t^2}.$$

Matched factorizations nearly collapse in this empirical clock.

Interpretation: the correct schedule coordinates may depend on the optimizer's effective homogeneity.

The theory becomes a re-parameterized functional scaling laws (Li et al. 2025)

Theory

$$R_\sigma(T) \asymp F(T) + \int_0^T \underbrace{\frac{F(u)+\sigma^2}{r(u)}}_{\text{injected at } u} \underbrace{k(T-u)}_{\text{survives to } T} \, \mathrm{d}u.$$

Modeling

$$\widehat{L}(T) = \underbrace{L_\infty}_{\text{floor}} + \underbrace{A_{\mathcal{F}}(1+T)^{-q_{\mathcal{F}}}}_{\text{unlearned signal}} + \int_0^T \frac{A_1(1+u)^{-q_{\mathcal{F}}} + \hat{\sigma}^2}{r(u)} \left(1 + c_{\mathcal{K}}(T-u)\right)^{-q_{\mathcal{K}}} \mathrm{d}u.$$

- $A_{\mathcal{F}}(1+T)^{-q_{\mathcal{F}}}$: the signal part that is unfinished to learn: modeling $F(t)$
- $1/r(u)$: schedule-dependent noise injection at time u
- $\left(1 + c_{\mathcal{K}}(T-u)\right)^{-q_{\mathcal{K}}}$: how much the noise can survival from u to T: modeling $k(T-u)$

Independent fits stay near $q_{\mathcal{K}} \approx 1$

Independent seven-parameter fits

Each dataset/model is trained and fitted independently.

Dataset / model	Tokens	$q_{\mathcal{K}}$	$q_{\mathcal{F}}$
OpenWebText / 124M	2.5B	1.017	0.374
FineWeb / 124M	2.5B	0.952	0.405
peS2o V2 / 124M	2.5B	0.995	0.365
OpenWebText / 300M	6.5B	0.989	0.291

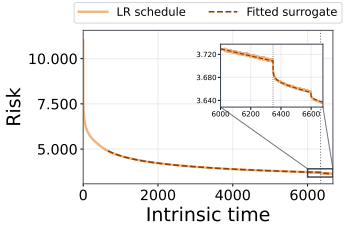
Finding. Across datasets and model scales, the fitted exponents satisfy $q_{\mathcal{K}} \approx 1$ and $q_{\mathcal{F}} < 1$, placing the tested LLMs near the $\mathrm{LM}_1/\mathrm{IM}_1$ boundary of the phase map.

A controlled test in 300M nanoGPT: Prediction

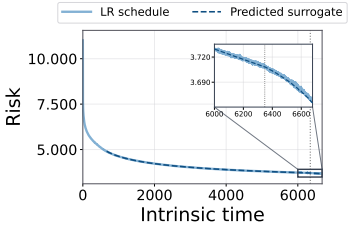
300M / plain SGD / OpenWebText; Risk = validation CE

Building on FSL (Li et al., 2025): surrogate structure and fit-then-transfer protocol.

Fit on 8-1-1



Predict WSD without refitting



$T_t = \sum_{s<t} \eta_s$: cumulative learning rate; dotted line: shared checkpoint.

No refitting. All fitted parameters come from 8-1-1; only the schedule changes.

Takeaway messages

Existence	F and K follow power laws exactly when their cumulative weighted low-spectrum masses have the corresponding scaling.
Modeling	The response exponents $(q_{\mathcal{F}}, q_{\mathcal{K}})$, intrinsic clock $T = \sum \eta$, and ratio path $r = B/\eta$ determine how clean learning competes with accumulated stochastic memory.
Prediction	These coordinates organize matched nanoGPT runs, and one fitted forcing-memory response predicts unseen schedules without refitting.

- Yichen Wang, Fanghui Liu, Yudong Chen. From Spectra to Joint Schedules in LLM Pre-training: 3+3+(2) Scaling-Law Regimes. https://arxiv.org/abs/2609.40148
- Code and configurations: https://github.com/yichenblue/spectra-to-schedules-in-pretraining

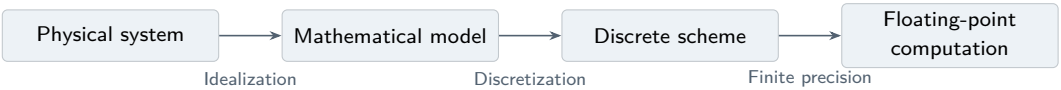
Deep Learning/LLMs as Technical Science

A familiar chain in scientific computing

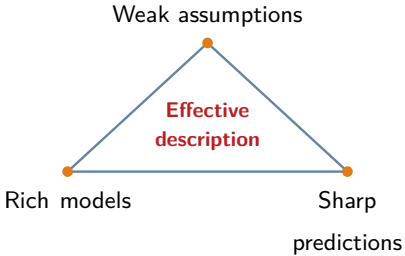


Deep Learning/LLMs as Technical Science

A familiar chain in scientific computing



A modeling trade-off



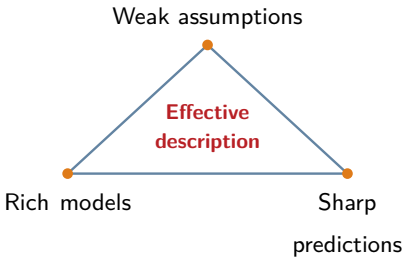
Preserve the relevant response.

Deep Learning/LLMs as Technical Science

A familiar chain in scientific computing



A modeling trade-off



Preserve the relevant response.

Core task: Problem formulation and modeling

Start from common problems and build effective descriptions.

Method: Analysis, computation, experiment

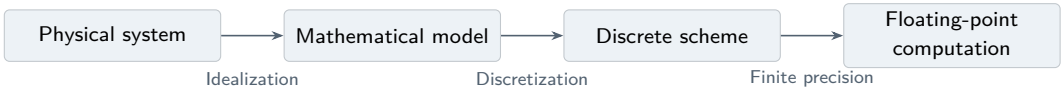
Human in a loop...

Goal

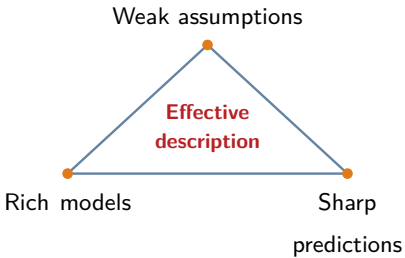
Explain mechanisms, characterize limits, develop predictions, and guide design

Deep Learning/LLMs as Technical Science

A familiar chain in scientific computing



A modeling trade-off



Preserve the relevant response.

Core task: Problem formulation and modeling

Start from common problems and build effective descriptions.

Method: Analysis, computation, experiment

Human in a loop...

Goal

Explain mechanisms, characterize limits, develop predictions, and guide design

The concern is not mathematics itself, but the disconnection between theory and practice.

Q & A

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References

1. Kaplan et al. (2020). *Scaling Laws for Neural Language Models*.
2. Hoffmann et al. (2022). *Training Compute-Optimal Large Language Models*.
3. Paquette et al. (2024). *4+3 Phases of Compute-Optimal Neural Scaling Laws*.
4. Li et al. (2025). *Functional Scaling Laws in Kernel Regression: Loss Dynamics and Learning Rate Schedules*.
5. Karpathy (2022). *nanoGPT*.
6. Jordan et al. (2024). *Muon: An Optimizer for Hidden Layers in Neural Networks*.
7. Hastie et al. (2019). *Surprises in high-dimensional ridgeless least squares interpolation*.
8. Nakkiran et al. (2019). *Deep double descent: Where bigger models and more data hurt*.
9. Neyshabur et al. (2015). *Norm-based capacity control in neural networks*.